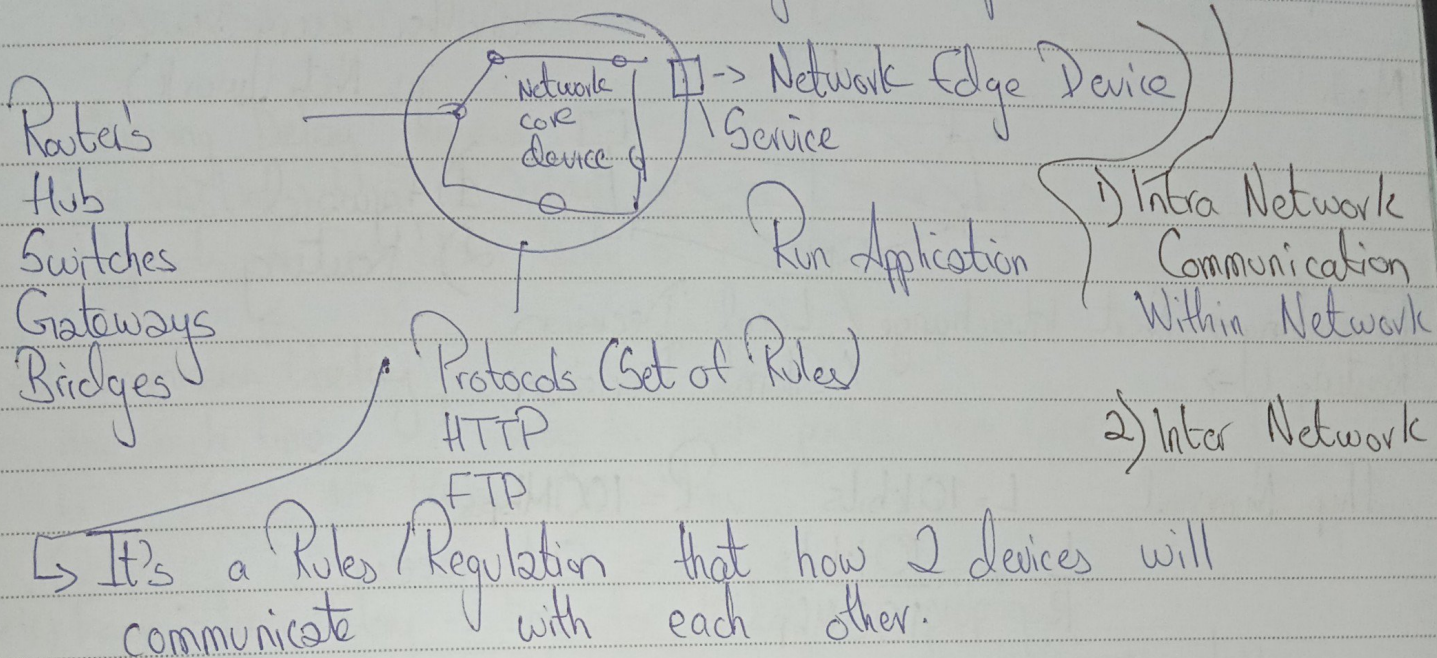


Computer Network

Date: 16/10/24

It is collection of devices work together to provide Communication.



Computer Networking Top Down Approach
Keith W. Ross
8th Edition.

17/10/24

Chapter 1

Network is connection of devices.
Protocol define format, order of message sent & recieved among network entities and actions taken on message transmission.

Host: Client & Servers
Server → In Data Centers

Access Network

Host Send Packets?

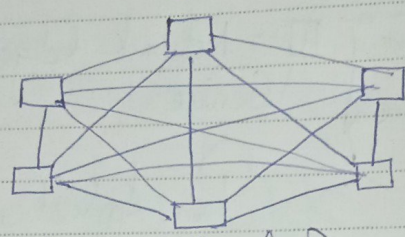
$\frac{L}{R}$

Packets Del

Date: 23/10/24

Network Core Topology $\rightarrow$ Bus, Star, Ring, Mesh (Every Device connected to another every device in Net (work))

Mesh

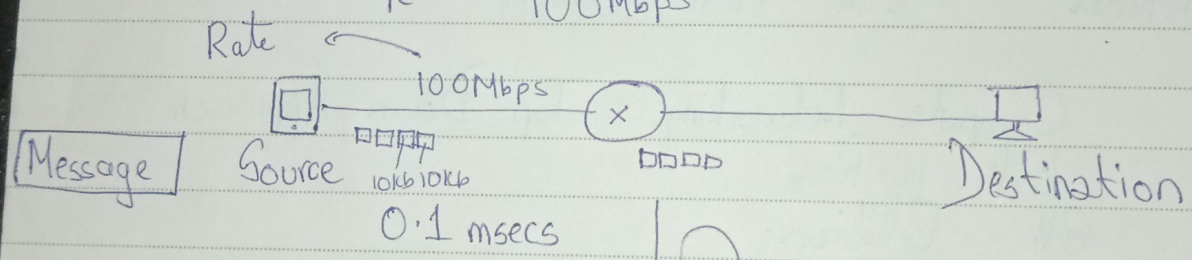


- 1) Forward
- 2) Routing

Forwarding $\rightarrow$ Next Interchange / Local Decision
 Routing $\rightarrow$ / Global Decision

1hop Numerical $L = 10\text{kbits}$ $R = 100\text{Mbps}$

$$\frac{L}{R} = \frac{10\text{kbits}}{100\text{Mbps}} = 0.1\text{ msec}$$



Router stores, won't send to destination until every packet is stored in Router.
 Packet Switch alternate is Circuit Switching.

29/10/24

Internet Structure "Network of Networks"
 ISP, IXP, Content Provider Network

Flat Structure vs Hierarchical Structure

Packet loss occurs when memory to hold queued packet fills up.
 Transmission Delay ($\frac{L}{R}$) Queue Delay.

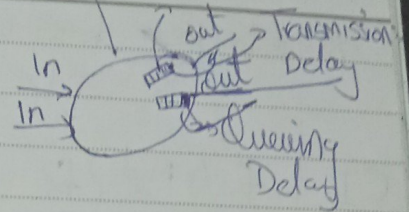
Sandhi

Caravan Analogy Quiz 2mins

Forward Table	
1011	1
0110	2

Date:

buffer



Quiz $\rightarrow$ 4 delays + What's difference between them.

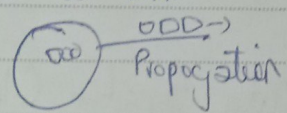
- 1) Nodal Processing delay
 - a) Checks bit error
 - b) Determine Output Link

- 2) Queueing Delay
 - a) Time Wasting at Output Link for Transmission.
 - b) Depends on Congestion level of Router.

- 3) Transmission Delay

How much time it take to push packet from Queue to Link.
 $\frac{L}{R}$ 1.5 Mbps / 100 Mbps

- 4) Propagation Delay $\rightarrow$ Time for Packet to propagate
 $\frac{d}{s} = \frac{dis}{speed}$



$$d_{nodal} = d_{proces} + d_{queue} + d_{trans} + d_{prop}$$

Caravan $\rightarrow 12 \times 10 = 120 \text{ secs} \rightarrow \frac{d}{s} = \frac{100}{100} \Rightarrow 1 \text{ hour}$

(Done in Wrong Place) Quiz #1

30/10/24

Next week Thursday

Descriptive, Inferential, Discrete, Continuous, Quantative, Population Sample.

Measure of Center Tendency

Population mein μ

Sample mein $\bar{x}$

AM, Median, Mode
Arithmetic

UnGroup
2, 3, 5, 7, 11
 $\bar{X} = \frac{\sum x}{n}$

Group Data

Class		
Class ^x	f	fx
2	2	4
3	9	27
5	1	5

Class Int		
CI ^x	f	fx
2-3	2	
4-5	1	
6-7	2	
8-9	1	

AM $\rightarrow$ Class CI,

some formulas.

not for Mean Mode

Date: _____

Arithmetic Mean formula for Grouped Data

$$\bar{X} = \frac{\sum fx}{\sum f}$$

$$\text{or } \bar{X} = \frac{\sum fx}{N}$$

CI	f	Mid $\bar{x}$	fx
2-3	2	2.5	5
4-5	3	4.5	13.5
6-7	4	6.5	26

Q Find out A. Mean

Step 1 $\rightarrow$ Find Mid Point

Step 2 $\rightarrow$ Calculate fx

$$\frac{\sum fx}{\sum f}$$

MEDIAN

Data Sensitive (Include Extreme Value) $\rightarrow$

Median

Ungroup

odd

$$\frac{n+1}{2}$$

even

$$\frac{n}{2} / \frac{n+2}{2}$$

Group

Class

CI

Find Out Median

2, 3, 2, 4, 2

Ascending Order

2, 2, 2, 3, 4

$$\frac{n+1}{2} = \frac{5+1}{2} = 3$$

$$\frac{2}{2}$$

2 is answer

Find Out Median

3, 2, 2, 3, 4, 2

Ascending Order

2, 2, 2, 3, 3, 4

$$\frac{2+3}{2} = 2.5$$

52 slide left nhi ha.

Introduction 1-52

Date: _____

Class	f	C.freq
2	4	4
3	3	7
4	2	9
5	4	13

$$\frac{\sum f}{2} = 6.5$$

6.5 k samne wali Class
Class will be Median.

Median mein Discrete k liye Class Boundary bananae gay.

CI	f	Class B	C F
2-3	3	1.5-3.5	3
4-5	2	3.5-5.5	5
6-7	4	5.5-7.5	9
8-9	5	7.5-9.5	14

$$\frac{\sum f}{2} = \frac{14}{2} = 7.$$

$$l + \frac{h}{f} \left(\frac{\sum f}{2} - Cf \right)$$

$$5.5 + \frac{2}{4} \left(\frac{14}{2} - 5 \right) \Rightarrow 6.5$$

Mode

Class I	f	
2-3	3	f_1
4-5	8	f_m
6-7	2	f_2

$$l + \frac{(f_m - f_1)}{(2f_m - f_1 - f_2)} \times h$$

$$4 + \frac{(8-3) \times 1}{(2 \times 8 - 3 - 2)}$$

Q1 → Delays + Calc
 Q2 → Concept Check (10 Q)
 Q3 →

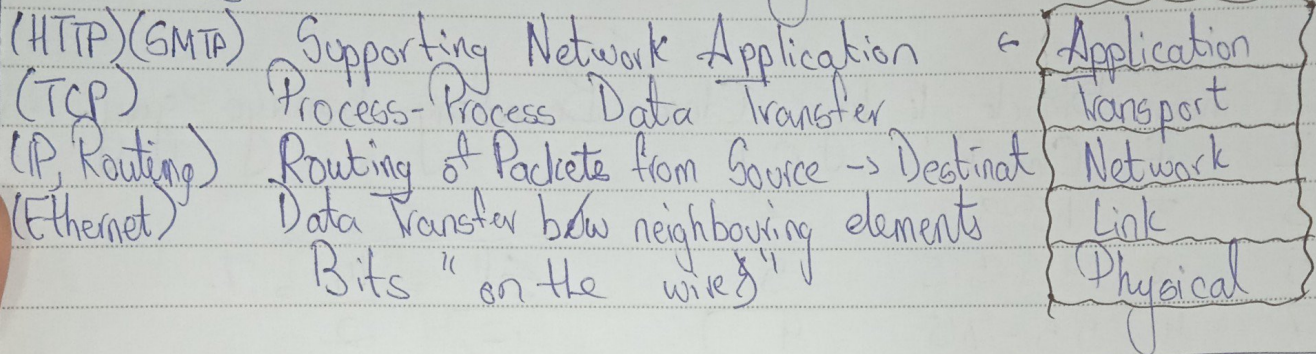
Date: 30/10/24

Traceroute, Packet Loss, Throughput, Security

TCP → Reliable

6/11/24

Layer Approach to design complex System, Ease Maintenance.



Encapsulation

Message + Header = Segment
 Network layer is datagram.
 Link is frame is the bit

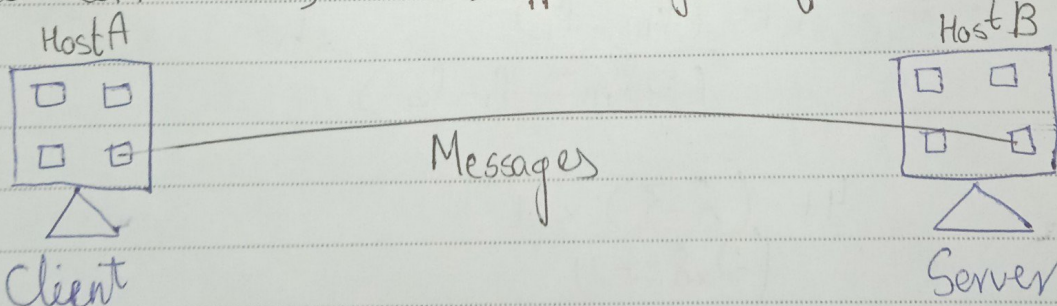
DNS Server
ssuet 192.168.0.1

7/11/24

Chap 2 → Client Server, Peer 2 Peer

13/11/24

Process Communication, Sockets, App Layer Define



Process Send/Receives messages to/from it's Socket.

Application Layer Protocol

- 1) Type of Message exchanged. (2) Message Syntax, Rules, Semantic
 Transport Service does
 Data Integrity, Timing, Throughput, Security

TCP

- 1) Transmission Control Protocol
- 2) Reliable (Acknow / Resend)
- 3) Flow Control (POV Receiver)
- 4) Congestion Control (bit Cong /
 (Goal for Networking))

- 5) Connection Oriented
 (Handshaking)

Doesn't Provide
 • Timing, Minimum Throughput
 guarantee, Security.

Receiver k liye acha sochna hai
 File Transfer / Email,
 Web Document, ~~Video~~ ~~Audio~~
 No delay guarantee

UDP

- 1) User Datagram Protocol
- 2) UnReliable Data Transfer
 (Fast, No Acknow & Resend)
- 3) No Flow Control
- 4) No Congestion Control, no timing
- 5) No Security, No Connection Setup
- 6) No Throughput Guarantee

Streaming Audio / Video

- 7) Unordered delivery

No delay guarantee

19/11/24

Non-Persistent HTTP

(One-Object sent
 over TCP)

RTT → time for
 small packet to
 travel from client
 to server & Back.

Persistent HTTP

} Multiple Object sent

Ch 25, 25

RTT → Round
 Trip Time

HTTP Stateless

Shubham?
mark

Date: _____

N. Pres HTTP Resp Time = $2RTT + \text{File Transmission Type}$.

2 type of HTTP $\rightarrow$ Request & Response

Request $\rightarrow$ (Methods)

POST (webpage often includes input)

GET (for sending data to server)

HEAD (Request Headers)

PUT (Upload new file to server)

Cookies $\rightarrow$

~~Web~~ Websites and client browser use cookies to maintain some state between transaction.

There are 4 Components. (2-33)

Maintaining User/Server State of Cookies.

How to keep State?

At protocol Endpoint and in messages.

Cookie & Privacy? Cookie permits sites to learn a lot about you on their site.

Cookies Used for

- Authorization

- Shopping Cart

- Recommendations

- User Session State (Email)

Web Cache

- its goal is to ~~del~~ satisfy a client request without involving origin Server

- act as both client & server.

- Reduce Response time for client

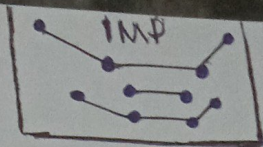
Email

SMTP

Mail Server, SMTP Protocol Alice bob $\rightarrow$ Push Protocol.

IMAP

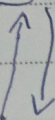
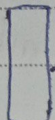
Shubham?



Date: _____

20/11/24
DNS → Domain Name System (Application Layer)
Distributed DB implemented in hierarchy of many
name servers. Lookup Table

DNS Server



Client

www.sscet.ac	198.168

Implemented as Application Layer Protocol

→ Hostname to IP Address Translation
Why not Centralized DNS?
Single Point of Failure, Traffic Volume, Distance Centralized DB
Maintenance

Hierarchical
Root → Top Level Domain → Authoritative (Organization Own)
Root → DNS → IP Address.

Queries → Top Level Domain → Recursive VS Iterative
Cache DNS Information

21/11/24

Chapter 3 Transport Layer
Transport layer vs Network layer
Ann & Bills Seen } 3-13 {
Multiplexing / Demultiplexing

How demultiplexing work?
Connectionless demultiplexing.

- Multiplexing & Demultiplexing based on segment, datagram, header field values
- Mul/Demul happens at all layer (Justify)
 UDP → Demul using destination port
 TCP → Demul uses Source, Destination, IP Address & Port Numbers.

Hub, Bridge 26/11/24

Network of Devices (1 Ques)

Switch, 2 layer Switch (Bridge with many ports)
 3 layer Switch Router

Given which Device when? Routers

Gateway

4/12/24

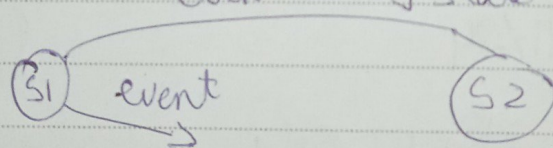
Principle of Reliable Data Transfer

4 function

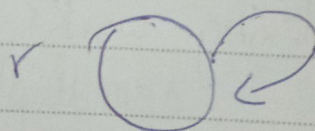
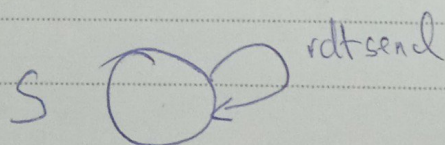
$rdt_send()$, $rdt_recv()$, $rdt_trcv()$, $deliver_data()$

event causing state transition

Unidirectional



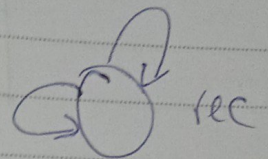
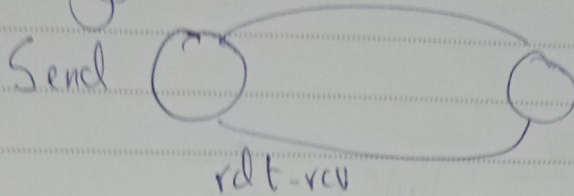
Develop rdt1.0



Rdt 2.0

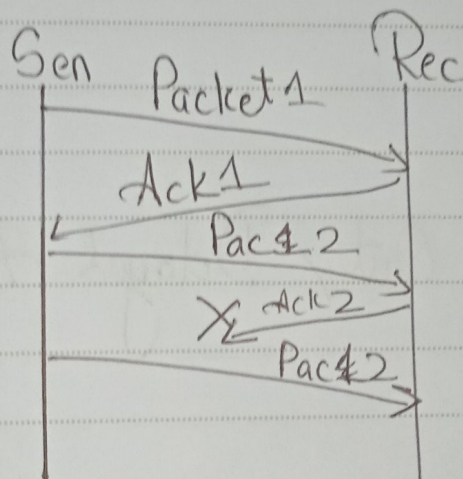
Date: _____

Acknowledgement
Negative Acknowledgement (Retransmit)
Stop & Wait



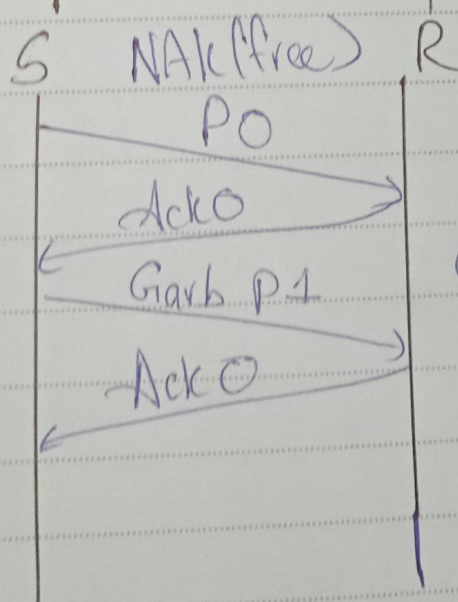
Stop & Wait Protocol

RDT 2.1



2 seq
Receiver expects

Rdt 2.2

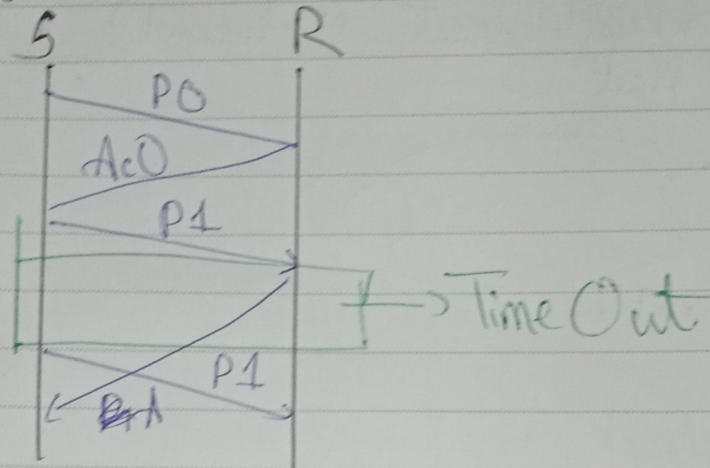


Garballed (flipped)

0,1

Date: _____

rdt3 Channels with error & loss
We add Timer



- 1) No Loss
- 2) Packet loss
- 3) Acknow loss
- 4) Premature Time Out / Delayed Acknow